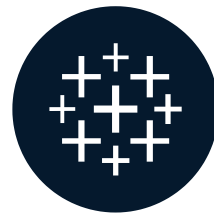


Introduction

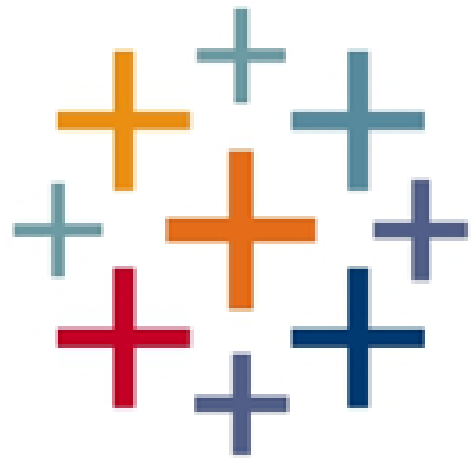
INTRODUCTION TO TABLEAU



Maarten Van den Broeck
Content Developer at DataCamp

What is Tableau?

- Data visualization tool
- Click, drag, drop
- Beautiful, interactive visualizations



+tableau

Why use Tableau?

- Accessible for a range of users
- Advanced analytical capabilities
- Flexible
- Intuitive
- Quick and robust prototyping
- Frame (business) questions
- Import and clean data
- Analyze and visualize data
- Drive business decisions
- Present insights

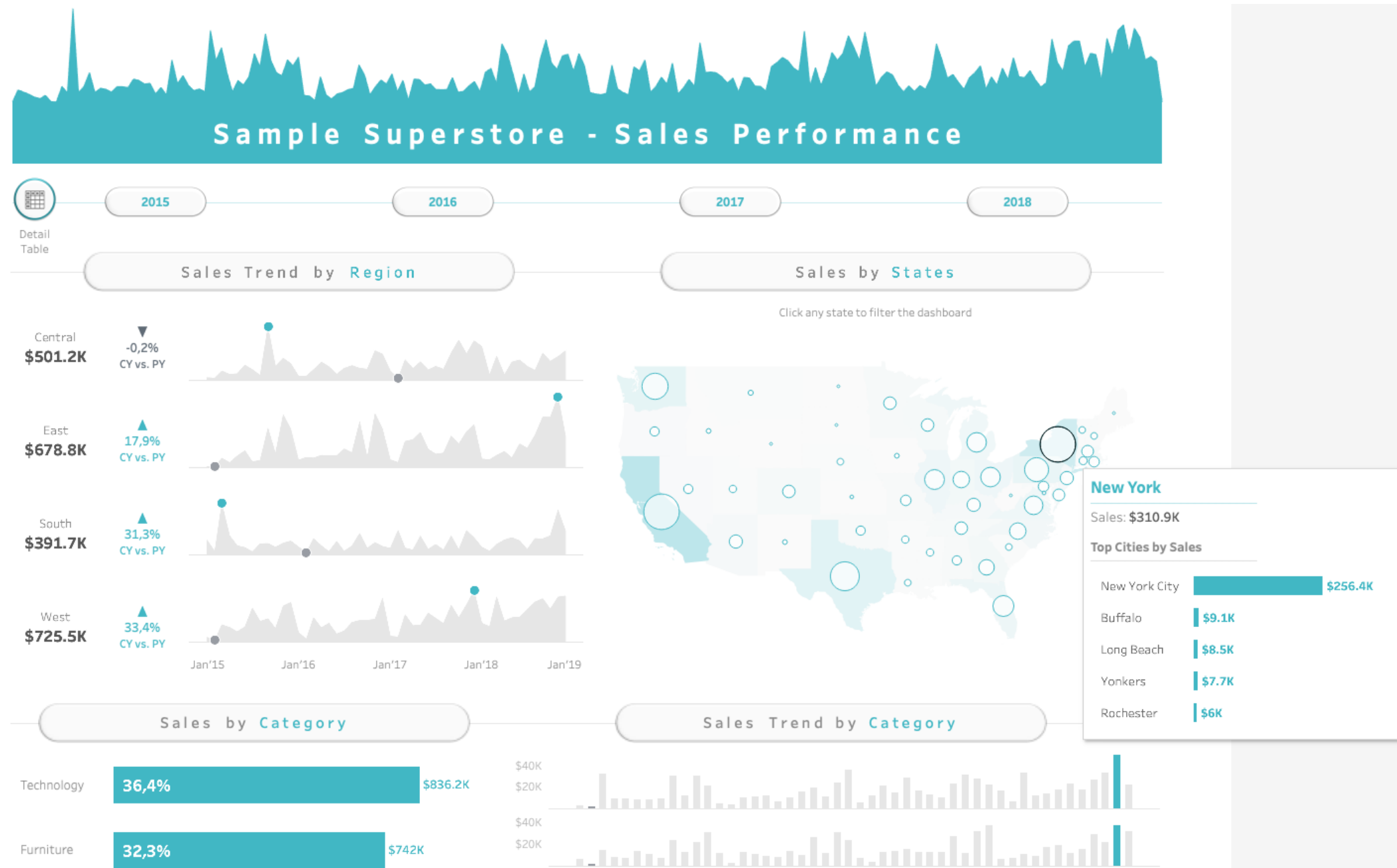


Who uses Tableau?

Roles

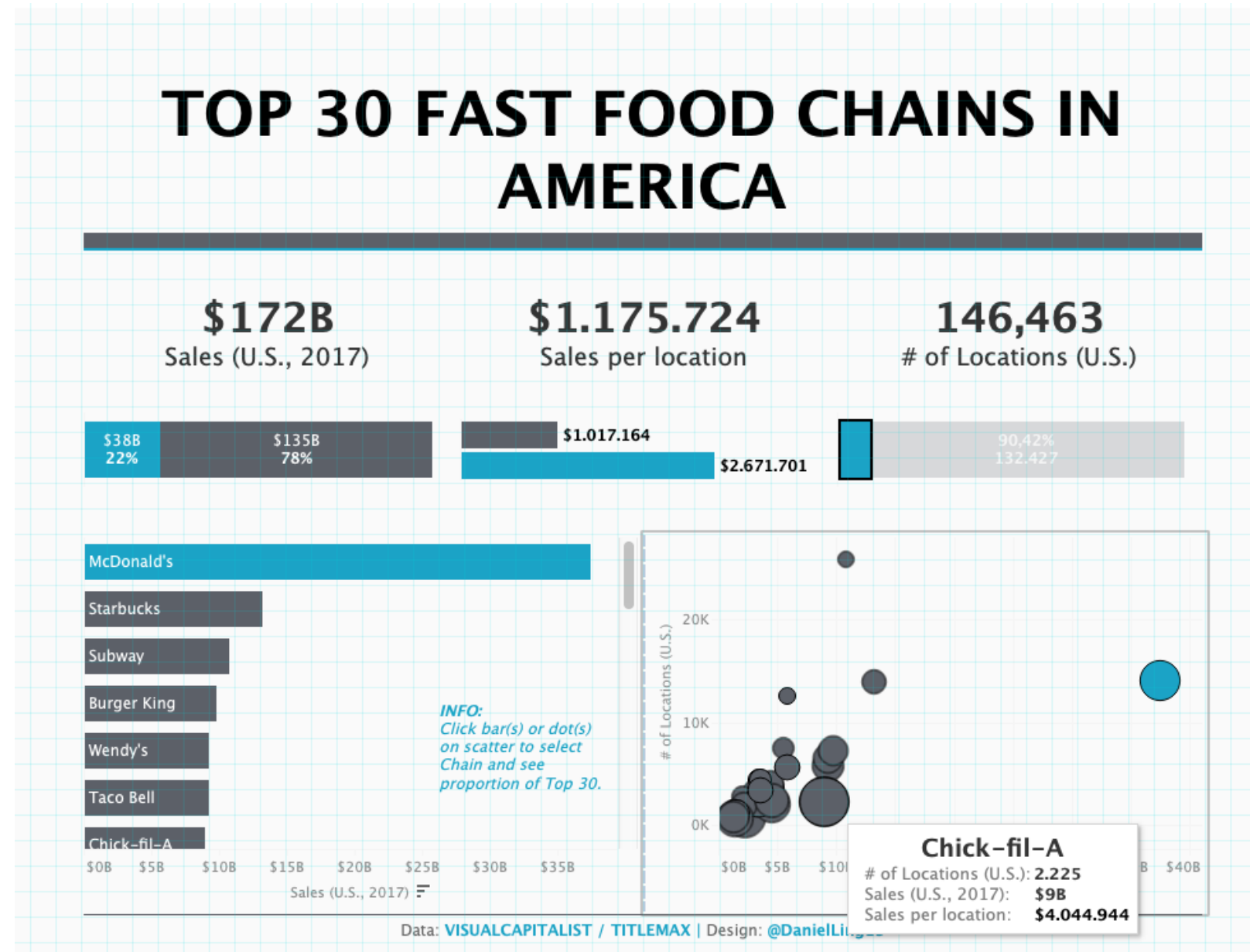
- Data analyst
- Business analyst
- Analytics consultant

Possibilities with Tableau



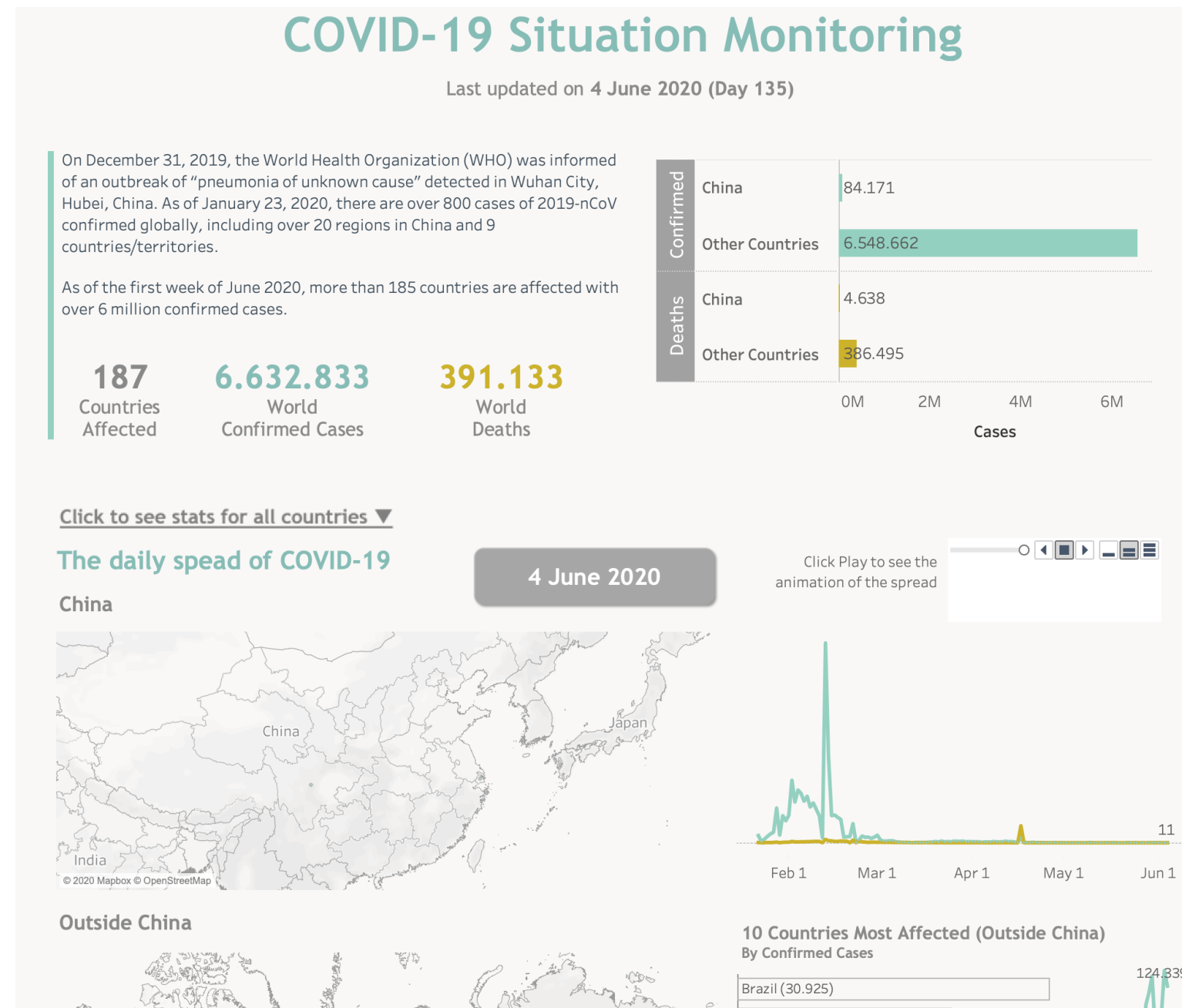
¹ Author: Pradeep Kumar G. Originally published on: Tableau Public

Possibilities with Tableau



¹ Author: Daniel Ling. Originally published on: Tableau Public

Possibilities with Tableau



¹ Author: Thi Ho. Originally published on: Tableau Public

Tableau versions

Tableau Desktop Public Edition

- Free
- All visualizations included
- Excel, csv, Google Sheets, web data
- 15 millions rows of data
- Publish locally¹ and online

¹ Since April 2024

Tableau versions

Tableau Desktop Public Edition

- Free
- All visualizations included
- Excel, csv, Google Sheets, web data
- 15 millions rows of data
- Publish locally¹ and online

Tableau Desktop

- Paid (license)
- All visualizations included
- All listed data sources
- Unlimited rows of data
- Publish locally and online

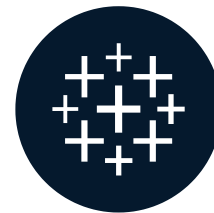
¹ Since April 2024

Let's practice!

INTRODUCTION TO TABLEAU

Connecting to data

INTRODUCTION TO TABLEAU



Maarten Van den Broeck

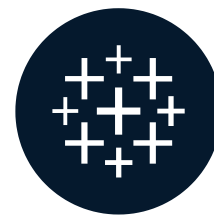
Content Developer at DataCamp

Let's practice!

INTRODUCTION TO TABLEAU

Navigating Tableau

INTRODUCTION TO TABLEAU



Maarten Van den Broeck

Content Developer at DataCamp

The image shows the Tableau Desktop interface. On the left, the 'Data' pane is active, displaying a table named 'san_francisco'. Below the table name is a search bar and a list of fields. The fields are categorized into 'Tables' and 'Measure Names'. The 'Tables' section includes 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', and 'Measure Names'. The 'Measure Names' section includes 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The top toolbar contains various icons for navigation, editing, and visualization. The main workspace is titled 'Sheet 1' and is currently empty, with a 'Drop field here' prompt visible. The bottom status bar shows 'Data Source' and 'Sheet 1'.

<

>

Standard

Show Me

DataAnalytics

san_francisco

Search

Tables

- # Id
- Abc Neighbourhood
- # Reviews per Month
- Abc Room type
- Abc Measure Names
- # Availability 2019
- # Days Occupied in 2018
- # F1
- 🌐 Latitude
- 🌐 Longitude
- # Minimum Nights
- # Number of Reviews
- # Price
- # san_francisco.csv (Count)
- # Measure Values

Pages

Columns

Rows

Filters

Marks

Automatic

ColorSizeTextDetailTooltip

Sheet 1

Drop field here

Drop field here

Data Source

Sheet 1

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields from the 'san_francisco' data source, including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic' and displays a grid of fields. The main view area is labeled 'Sheet 1' and contains a large empty space with the text 'Drop field here'.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields from the 'san_francisco' data source, including 'Reviews per Month', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic' and is empty. The main view area is labeled 'Sheet 1' and contains the text 'Drop field here'.

The image shows the Tableau Desktop interface. On the left is the 'Data' pane with a search bar and a list of fields. The 'Columns' and 'Rows' shelves are at the top. The 'Marks' card is in the center. The main view area is labeled 'Sheet 1' and contains a large empty space with the text 'Drop field here'. The 'Data' pane lists various fields including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic'. The main view area is labeled 'Sheet 1' and contains a large empty space with the text 'Drop field here'.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields from the 'san_francisco' data source, including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic'. The main view area is labeled 'Sheet 1' and contains three 'Drop field here' prompts.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields for the 'san_francisco' data source, including 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic' and is empty. The main view area is labeled 'Sheet 1' and contains three 'Drop field here' prompts.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' shelf on the right. The 'Data' pane lists fields from the 'san_francisco' data source, including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Reviews per Month' field is highlighted in green. A context menu is open over the 'Reviews per Month' field, showing options: 'Add to Sheet', 'Duplicate', 'Rename', 'Hide', 'Create', 'Convert to Discrete', 'Convert to Measure' (highlighted in blue), 'Change Data Type', 'Geographic Role', 'Default Properties', 'Group by', 'Folders', 'Hierarchy', 'Replace References...', and 'Describe...'. The main workspace area is labeled 'Sheet 1' and contains two 'Drop field here' prompts. The bottom status bar shows 'Data Source' and 'Sheet 1'.

The screenshot shows the Tableau Desktop interface. On the left, the 'Data' pane displays a list of fields from the 'san_francisco' data source. The field 'Reviews per Month' is highlighted. A right-click context menu is open over this field, showing various actions. The 'Convert to Discrete' option is currently selected and highlighted in blue. The main workspace, 'Sheet 1', is empty, showing a grid with 'Drop field here' prompts. The top toolbar contains standard Tableau navigation and editing tools. The bottom status bar shows 'Data Source' and 'Sheet 1'.

Data | Analytics

san_francisco

Search

Tables

- # Id
- Abc Neighbourhood
- # **Reviews per Month**
- Abc Room type
- Abc Measure Names
- # Availability 2019
- # Days Occupied in 2018
- # F1
- Latitude
- Longitude
- # Minimum Nights
- # Number of Reviews
- # Price
- # san_francisco.csv (Count)
- # Measure Values

Columns

Rows

Sheet 1

Drop field here

Drop field here

Drop field here

Context Menu:

- Add to Sheet
- Duplicate
- Rename
- Hide
- Create
- Convert to Discrete**
- Convert to Measure
- Change Data Type
- Geographic Role
- Default Properties
- Group by
- Folders
- Hierarchy
- Replace References...
- Describe...

Data Source | Sheet 1

Data roles in Tableau

Discrete dimension

- Common, colored in blue
- Finite amount of values
- Can't be aggregated
- *E.g.* eye color, sex

Continuous measure

- Common, colored in green
- Infinite amount of values
- Can be aggregated
- *E.g.* height, weight

Data roles in Tableau

Discrete dimension

- Common, colored in blue
- Finite amount of values
- Can't be aggregated
- *E.g.* eye color, sex

Continuous dimension

- Not common, colored in green
- Infinite amount of values
- Can't be aggregated
- *E.g.* date

Discrete measure

- Not common, colored in blue
- Finite amount of values
- Can be aggregated
- *E.g.* shoe size, age

Continuous measure

- Common, colored in green
- Infinite amount of values
- Can be aggregated
- *E.g.* height, weight

Segmenting with dimensions

- Dimensions and measures affect visualizations differently:
 - Dimensions are used to **segment** data
 - Measures can be aggregated
- **Segmenting**: grouping similar data together
 - *E.g.* average price per room type

Tableau interface showing the 'Data' pane on the left, the 'Columns' shelf with 'Neighbourhood' in the 'Rows' shelf, and the 'Marks' shelf. The main view is 'Sheet 1' with a blank canvas and 'Drop field here' prompts. The bottom status bar shows 'Data Source' and 'Sheet 1'.

The interface includes a top toolbar with navigation and formatting icons, a 'Data' pane on the left with a search bar and a list of fields, and a central workspace for building visualizations. The 'Columns' shelf is empty, and the 'Rows' shelf contains the 'Neighbourhood' field. The 'Marks' shelf is empty, and the 'Filters' shelf is empty. The main view is 'Sheet 1' with a blank canvas and 'Drop field here' prompts.

Data | Analytics

san_francisco

Search

Tables

- # Id
- Abc Neighbourhood
- # Reviews per Month
- Abc Room type
- Abc Measure Names
- # Availability 2019
- # Days Occupied in 2018
- # F1
- Latitude
- Longitude
- # Minimum Nights
- # Number of Reviews
- # Price
- # san_francisco.csv (Count)
- # Measure Values

Columns

Rows | Neighbourhood

Filters

Marks

Automatic

Color | Size | Text

Detail | Tooltip

Sheet 1

Drop field here

Drop field here

Drop field here

Data Source | Sheet 1

The image shows the Tableau Desktop interface. At the top is a toolbar with navigation and editing icons. Below the toolbar is a header bar with tabs for 'Data' and 'Analytics', and a dropdown menu showing 'san_francisco'. On the left is a sidebar with a search bar and a list of tables under the heading 'Tables'. The tables listed are: 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. In the center is a 'Marks' card with a dropdown menu set to 'Automatic' and six icons: Color, Size, Text, Detail, Tooltip, and a fourth icon. To the right of the Marks card is a large area labeled 'Sheet 1' with a grid. The grid has a header row and a header column. The text 'Drop field here' is visible in the top-right and bottom-right cells of the grid. At the bottom is a footer bar with tabs for 'Data Source' and 'Sheet 1', and several icons for view and layout settings.

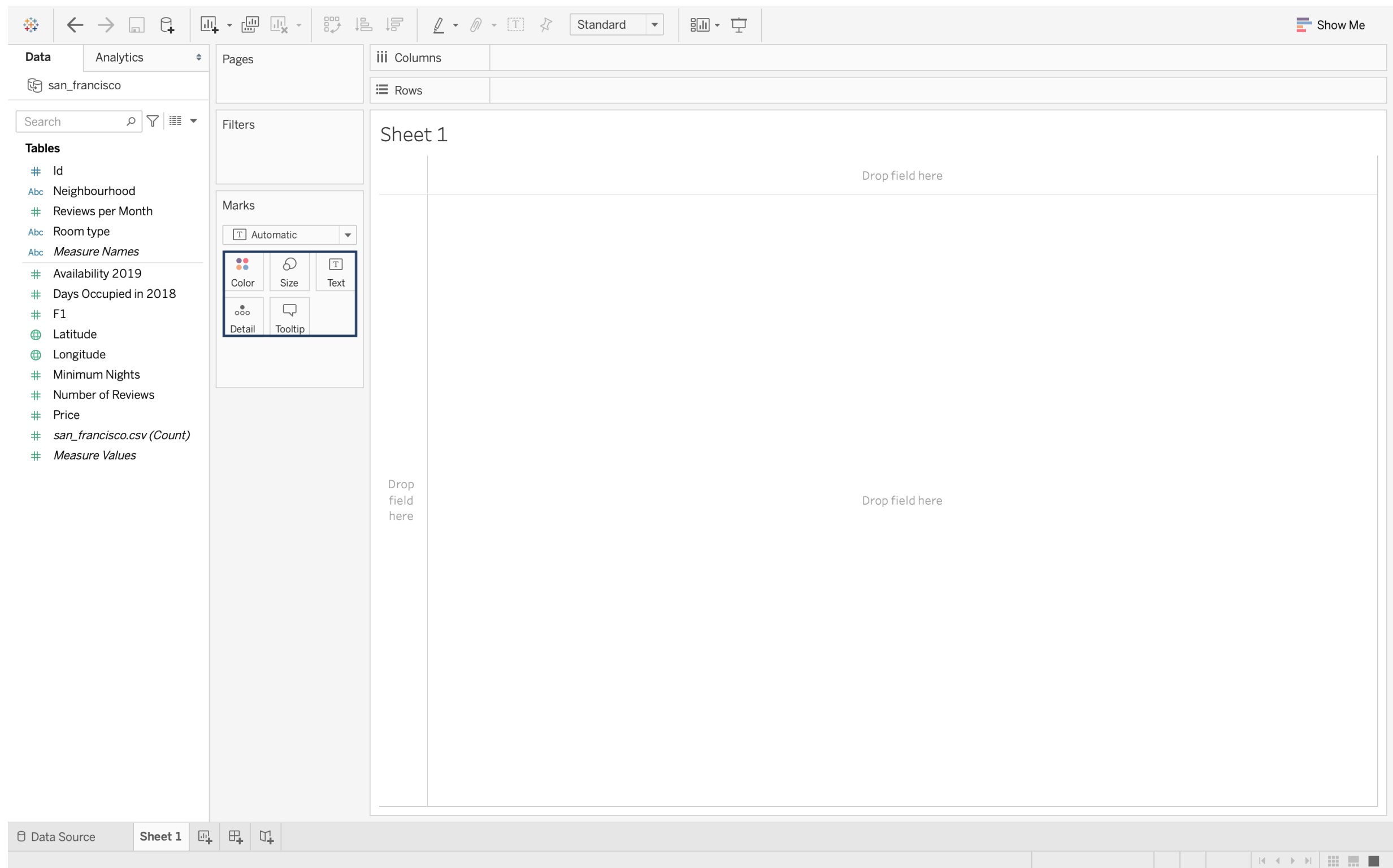
The image shows the Tableau Desktop interface. At the top is a toolbar with various icons for navigation and formatting. Below the toolbar is a sidebar on the left with a 'Data' pane. The 'Data' pane shows a list of tables and fields: 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The main workspace is titled 'Sheet 1'. It contains a large rectangular area with a placeholder text 'Drop field here'. The 'Columns' shelf at the top of the workspace is empty. The 'Rows' shelf is also empty. The 'Marks' card is set to 'Automatic' and has a placeholder text 'Drop field here'.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields from the 'san_francisco' data source, including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic' and shows a vertical bar chart. The main view area is labeled 'Sheet 1' and contains a large empty space with the text 'Drop field here'.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields for 'san_francisco', including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic' and includes options for Color, Size, Text, Detail, and Tooltip. The main view area is labeled 'Sheet 1' and contains the text 'Drop field here'.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields for 'san_francisco', including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' card is set to 'Automatic' and includes options for Color, Size, Text, Detail, and Tooltip. The main view area is labeled 'Sheet 1' and contains a large empty space with the text 'Drop field here'.

Tableau interface showing the 'Data' pane on the left, the 'Columns' and 'Rows' shelves at the top, and the 'Marks' card in the center. The 'Data' pane lists fields for 'san_francisco', including 'Id', 'Neighbourhood', 'Reviews per Month', 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The 'Marks' card is currently set to 'Automatic' and includes options for Color, Size, Text, Detail, and Tooltip. The main view area is labeled 'Sheet 1' and contains a grid with 'Drop field here' prompts.



The image shows the Tableau Desktop interface. On the left, the 'Data' pane displays a list of tables and fields. The 'Columns' shelf at the top contains 'Reviews per Month' and 'Neighbourhood'. The 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. The main view area shows a bar chart with 'Reviews per Month' on the vertical axis and 'Neighbourhood' on the horizontal axis. The chart displays bars for various neighborhoods, with 'Downtown' having the highest number of reviews. The interface includes a top toolbar with navigation and editing tools, a bottom status bar showing 'Data Source' and 'Sheet 1', and a right sidebar with a 'Show Me' button.

The image shows a data visualization tool interface. On the left is a sidebar with a 'Data' tab and a search bar. Below the search bar is a list of tables: 'Id', 'Neighbourhood', 'Reviews per Month' (highlighted in green), 'Room type', 'Measure Names', 'Availability 2019', 'Days Occupied in 2018', 'F1', 'Latitude', 'Longitude', 'Minimum Nights', 'Number of Reviews', 'Price', 'san_francisco.csv (Count)', and 'Measure Values'. The main workspace is divided into 'Columns' and 'Rows' sections at the top, and a large 'Sheet 1' area below. The 'Columns' and 'Rows' sections are empty. The 'Sheet 1' area contains a large empty space with the text 'Drop field here' in the top right and bottom right corners. On the left side of the 'Sheet 1' area, there is a vertical line and the text 'Drop field here'. At the bottom of the interface is a status bar with a 'Data Source' tab, a 'Sheet 1' tab, and several icons for data manipulation and visualization. The top of the interface features a toolbar with various icons for navigation, editing, and visualization, including a 'Standard' dropdown menu.

Our business question

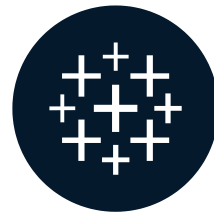
Which neighborhood and room type has the highest price in New York?

Let's practice!

INTRODUCTION TO TABLEAU

A tour of the interface

INTRODUCTION TO TABLEAU



Hadrien Lacroix

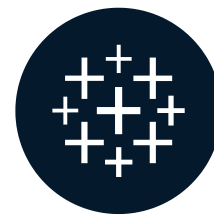
Content Developer at DataCamp

Let's practice!

INTRODUCTION TO TABLEAU

How to create visualizations in Tableau

INTRODUCTION TO TABLEAU



Maarten Van den Broeck
Content Developer at DataCamp

Let's practice!

INTRODUCTION TO TABLEAU